

Introduction to Databases

Introduction

Floris Geerts

Course Organization

Lectures and Exercises

- Theory lectures:

Wednesdays: 10:45-12:45 M.G.005

Slides on blackboard

Recording from a couple of years ago

<https://msevp.uantwerpen.be/Mediasite/Channel/inleidingdatabases/>

- Exercise and Q&A sessions :

Tuesdays: 10:45 – 12:45 M.G.025 (pc lab)

Lecturer and teaching assistant

Prof. Dr. Floris Geerts (Theory)

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Kasper Engelen, Tamara Cucumides (Exercises)

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- Appointments: send an email to arrange a meeting

Learning Objectives

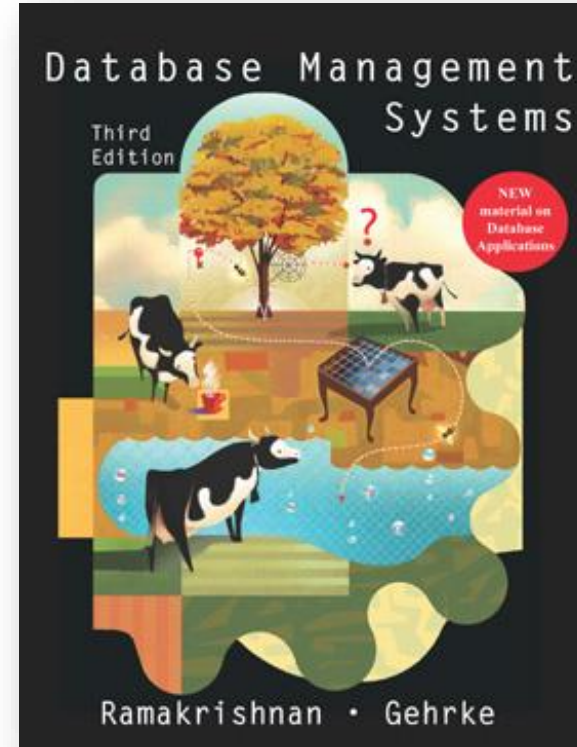
- Design a conceptual model for a given data description with the entity-relationship (ER) modelling language.
- Translate a conceptual ER model into a logical relational model including tables, primary keys, foreign key references, and constraints.
- Formulate database queries in the languages SQL and Relational Algebra.
- Understand and apply the principles of correct database design; this includes decomposing a given schema with functional dependencies into 3NF and BCNF.
- Explain techniques for efficient data retrieval such as indices and transaction management.
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Examination

- The final exam is going to be a **written, closed-book exam** that addresses both of the lecture's main components:
 - An **exercise part** with questions following the exercise sessions.
 - A **theory part** with questions about the lecture content.
- Typical question:
 - Given a conceptual model (ER diagram)
 - Translate this model into a logical schema
 - Formulate queries in one of the query languages
 - Find minimal keys, Check lossless/dependency preservation of decompositions, given BCNF, 3NF
 - Conflict serializability of schedules, Aries recovery method

Course Material

- Slides
- Recordings
- Book (mostly for extra background)



R. Ramakrishnan, J. Gehrke:
Database Management Systems,
2nd Ed., McGraw-Hill, 2000

Motivation: Databases Are Everywhere

- Information needs to be stored, used and manipulated in many types of applications:
 - administrative applications: banking, airline reservations and schedules, student administration, retail (customers, product recommendations, purchases, order tracking, bookkeeping), ...
 - document-oriented applications: newspapers, news sites, (digital) libraries, websites, search engines
 - technical applications: air traffic control, airplane control, motor management, ...



Motivation: Why Do We Need Databases?

- In many systems the data is central
- Data is shared by many programs
 - Student registration system
 - Grade registration system
 - Tuition fees administration
 - Mailings to all students
- Many functionalities are shared among data-intensive applications

Why can't we just use file systems?

- Data redundancy and inconsistency
 - Multiple file formats, duplication of information in different files
- Need to write a new program to carry out each new task
- Concurrent users
- Recovery after crashes

Summary: Motivation

- Database systems: highly specialized and optimized components
 - Data independence (changes to physical storage, or structure of data, hidden from software layers above)
 - Efficient data access
 - Data integrity and security
 - Data administration
 - Concurrent access and crash recovery
 - Reduced application development cycle
- In this course we will learn about the most common type of database system: the **relational database**